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Authors: Rick Moore

12.1. SOLID CERAMIC VERSUS FREQUENCY

Table 12.1 shows the parametric fits of permittivity for various ceramics. All ceramic samples were machined from large parts and measured in circular waveguide or transmission line cavity. The fitted functional form is shown at the beginning of the table. Functions are those causal functions that were derived in **Chaps. 2** and **3** for dielectric and magnetic media. When available, additional data are shown; these may include density, measured frequency span serving as basis for fits, tabulated information, and chemical and/or physical composition. In the tables exponentials are often shown as bE—which means $b \cdot 10^{-a}$. Thus $-2E-004-7.6E-003j$ is equal to $-0.0002-j0.0076$. Typical fits (showing measured and fitted functions) are presented in **Fig. 12.1** for BeO and Al_2O_3 , **Fig. 12.2**. **Figure 12.2** illustrates how fitting can smooth anomalous measured data that often occur with low-loss materials.

Table 12.1 *Ceramics versus Frequency: Data Are Real and Imaginary*

Sample ID							
$\epsilon(f) = B + 2Cf^D + G[1 - J^*(f-H)^2 - J2If_0^{-1}]$							
B	C	D	G	H	I	J	
3D Quartz–alumina silica nitride 2.16 g/cc (8–40 GHz)							
0.9921–	1.0147–	–8.79E–002+	0.9732–	3E–003+	–2E–004–	–2.06E–004+	
1E–004j	4E–004j	2E–003j	2E–004j	4E–004j	7.6E–003j	1.03E–005j	
Alumina 99.5% dense (1–250 GHz)							
2.399+	2.399+	1.22E–004–	2.40+	–6.93E–006	1.33E–005–	–5.62E–008–	
1E–004j	1E–004j	1.93E–005j	1E–004j	–3.75E–006j	2.31E–006j	3.26E–008j	
Alumina 99.9% dense 3.86–3.90 g/cc (1–300 GHz)							
2.3945+	2.3985+	2.43E–002–	2.39+	5E–004	1E–004+	2.30E–006–	
3.3E–003j	3.2E–03j	1E–003j	3.4E–003j		7E–004j	2.94E–007j	
Alumina 96–97% dense, 3.71 g/cc (5–250 GHz)							
2.4313+	2.4285+	1.12E–	2.2536–	4E–004	1.3E–003j	6.48E–006–	
1.5E–003j	1.4E–003j	002–1E–004j	2.9E–003j			6.25E–008j	
SRM709 (0.01–18 GHz), Lead oxide glass							
4.0907–	4.0907–	2E–004–	4.0902–	1E–004–	7.4E–003+	5E–004–	

Sample ID						
$\varepsilon(f) = B + 2Cf^D + G[1 - J^*(f-H)^2 - J_2/f_0^{-1}]$						
B	C	D	G	H	I	J
1.74E-002j	1.74E-002j	3.1E-003j	1.68E-002j	1.4E-003j	4E-003j	6E-004j
Mullite 97% dense (2–35 GHz), 3 Al ₂ O ₃ · 2SiO ₂						
1.6387+	1.6387+	1E-004–	1.6387+	–2.16E-005–	1.14E-005–	–4.63E-007+
7E-004j	7E-004j	1E-004j	7E-004j	1.42E-005j	3.09E-005j	8.63E-007j
Magnesium oxide (MgO) (2–35 GHz)						
2.3743+	2.3743+	2.4E-003	2.3743+	4E-004–	5E-004j	1.75E-005+
3E-004j	3E-004j		3E-004j	1E-004j		5.91E-007j
Slip-cast silica, 2.05 g/cc (2–35 GHz)						
0.8174+	0.8174+	1.2E-003+	0.8174+	–2E-004–	–1.21E-005+	3.62E-006+
1E-004j	1E-004j	1E-004j	1E-004j	1E-004j	3.56E-004j	1.33E-006j
Shuttle tile LI2200 (30–100 GHz)						
0.3371+	0.3371+	–2.6E-003	0.3374+	5E-004–	1E-004+	1.53E-005
1.2E-003j	1.2E-003j	–6E-004j	1.2E-003j	2E-004j	7E-004j	–2.49E-007j
Shuttle tile FRIC12 (3–100 GHz)						
0.2741+	0.2741+	4E-002–	0.2755+	–3.1E-003+	1E-004	–1.39E-005
6E-004j	6E-004j	8E-004j	5E-004j	1E-004j	–3E-004j	–3.60E-007j
Beryllium oxide (BeO) (0.2–250 GHz)						
1.6503+	1.6503+	–2.3E-003	1.6502+	1.042E-005	2.546E-006	–1.84E-006
1E-004j	1E-004j		1E-004j	–1.02E-006j	–3.57E-004j	–1.09E-008j
Boron nitride; 2.28 g/cc (1–40 GHz) (Nominal ε_r = 4.08, Accumet Engineering)						
1.1255+	1.1255+	1E-004+	1.1255+	1E-004–	1E-004	5.02E-006+
2E-004j	2E-004j	6E-004j	2E-004j	1E-004j	–1.1E-003j	5.04E-006j
Magnesium calcium titanate 30 (8–50 GHz) (Nominal ε_r = 30)						
–0.2585+	8.1811+	0.4159 +	0.5493+	–2.4E-003+	–2.16E-002j	–5E-004

Sample ID						
$\epsilon(f) = B + 2Cf^D + G[1 - J^*(f-H)^2 - J_2/f_3^{-1}]$						
B	C	D	G	H	I	J
2.7E-003j	0.5195j	0.3298j	8E-004j	1E-004j		
SRM 709 Lead-oxide glass (.01-18 GHz)						
4.0905-	4.0905-	2E-004-	4.0905-	1E-004-	7.4E-003+	5E-004-
1.73E-002j	1.73E-002j	3.1E-003j	1.75E-002j	1.4E-003j	4E-003j	6E-004j
SRM 710a Sodalime glass (.01-18 GHz)						
1.7687-	1.7687-	-2.4E-003	1.7687-	-4E-004+	1.7E-003	-1E-004-
4E-004j	4E-004j	-3E-004j	4E-004j	1.5E-003j	-6E-004j	1E-004j
PyroCeram (2-40 GHz)						
1.4171+	1.4171+	-2E-004	1.4171+	-1.94E-007	1.7437E-	-2.36E-006-
3E-004j	3E-004j		3E-004j	-1.37E-008j	005-1.5427E-004j	3.28E-008j
Sapphire wafer (.1" thick, 2" dia.) #1 of two perpendicular E field directions (80-100 GHz)						
2.8008+	2.7989+	-7.77E-002+	2.8204+	4E-003-	-6E-004- 4.6E-	-4.93E-005+
3.29E-002j	3.9-002j	2.5E-002j	2.54E-002j	1.2E-003j	003j	5.30E-006j
Sapphire wafer (.1" thick, 2" dia.) #2 of two perpendicular E field directions (80-100 GHz)						
2.4304+	2.4285+	2.47E-002+	2.4585+	4.1E-003+	-4.9E-003+	2.72E-005+
4.26E-002j	4.14E-002j	2.37E-002j	4.76E-002j	3.2E-003j	2.7E-003j	4.04E-005j
Silicon nitride 3.2-3.3 g/cc (2-35 GHz)						
1.3598+	1.36+	1.15E-002+	1.3599+	1E-003-	1.4E-003j	5.96E-005+
2.4E-003j	2.5E-003j	1E-004j	2.5E-003j	8E-004j		4.61E-006j
Fused silica-glass (2-40 GHz) Dynasil 4000 (generic)						
0.9688+	0.9688+	-1.2E-003+	0.9688+	2E-004+	-1.2161E-005	-7.15E-006+
1E-004j	1E-004j	1E-004j	1E-004j	2E-004j	-4.6692E-005j	6.33E-007j
Fused silica glass Dynasil 4000, 2.16-2.2 g/cc (0.1-100 GHz)						
0.9469+	0.9469+	-8.1E-003	0.9468+	-3E-004	-1.9E-003j	-3.35E-005
1E-004j	1E-004j		1E-004j			-7.4E-008j

Sample ID						
$\epsilon(f) = B + 2Cf^D + G[1 - J^*(f-H)^2 - J_2/f_1^{-1}]$						
B	C	D	G	H	I	J
Spinel magnesium aluminum oxide MgAl ₂ O ₄ (2–35 GHz)						
2.0614+	2.0615+	9.8E-003+	2.0614+	2E-003	2.2E-003j	8.64E-005+
4E-004j	4E-004j	1E-004j	4E-004j			1.07E-006j
Magnesium titanate 16 (0.01–18 GHz) (nominal $\epsilon_r = 16$)						
4.2136+	4.2131+	9.4E-003+	4.2135+	7.3E-003+	8.5E-003+	4E-004-
9.9E-003j	9.9E-003j	5.5E-003j	9.9E-003j	7E-004j	4.6E-003j	1.1E-003j
Barium titanate 38, 4.41–4.48 g/cc (8–50 GHz) (nominal $\epsilon_r = 38$)						
7.5015+	6.3603+	0.1914	8.0033-	-6.1E-003+	3E-004	-1.6E-003
1.52E-002j	7.3E-003j		4.36E-002j	2E-004j	-1.74E-002j	
Cordierite Mg ₂ Al ₄ Si ₅ O ₁₈ 4, 2.3–2.5 g/cc (8–50 GHz)						
1.2293+	1.5892+	-0.2778	1.6774+	-6.7E-003	5E-004	4.43E-005- 2.35E-
1.19E-002j	2.3E-003j	-3.36E-002j	0.1004j	-3.5E-003j	-2.2E-003j	005j
Magnesium aluminum titanate 11, 3.3–3.5 g/cc (8–50 GHz) (nominal $\epsilon_r = 11$)						
2.3079-	2.5009+	0.1472-	1.7366+	1.14E-002-	1.2E-002-	-3E-004-
3.25E-002j	0.2874j	0.1533j	0.8388j	3.19E-002j	1.55E-002j	3E-004j
Stycast 2662 (10–60 GHz) 1.385 g/cc (nominal 1.44 g/cc, $\epsilon_r = 3.5$ –3.61 MHz)						
0.4099+	0.8644	0.1859-	0.8048+	3.2E-003	1.03E-002-	-1E-004-
7.1E-003j	-5.51E-002j	8.5E-002j	0.4175j	-1.82E-002j	9.3E-003j	2E-004j
Corning 7940 fused silica, 2.201 g/cc (nominal $\epsilon_r = 3.8$ at MHz) (10–100 GHz)						
0.9389-	0.9413-	2.61E-002+	0.9381-	1.7E-003	-1E-004+	3.50E-005+
3E-004j	2E-004j	8E-004j	5E-004j		2.6E-003j	1.31E-006j
Corning 7957 fused silica, 2.20 g/cc (nominal $\epsilon_r = 3.8$ at 1 MHz) (27–60 GHz)						
0.9213	0.936-	4.7E-002+	0.9583-	1E-004+	-6.4E-003+	3.63E-005+
-3.07E-002j	2.03E-002j	5.24E-002j	7.7E-002j	6.4E-003j	1.9E-003j	6.44E-005j

Figure 12.1 Measured and model fit of BeO permittivity.

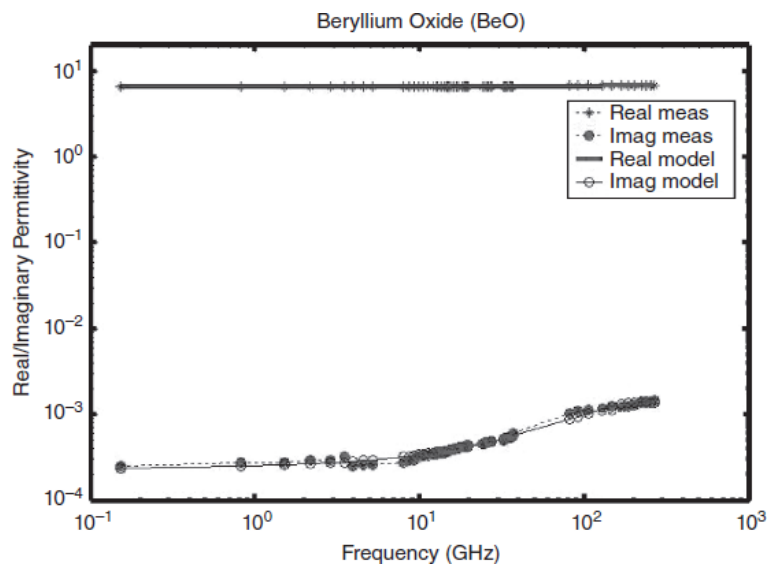
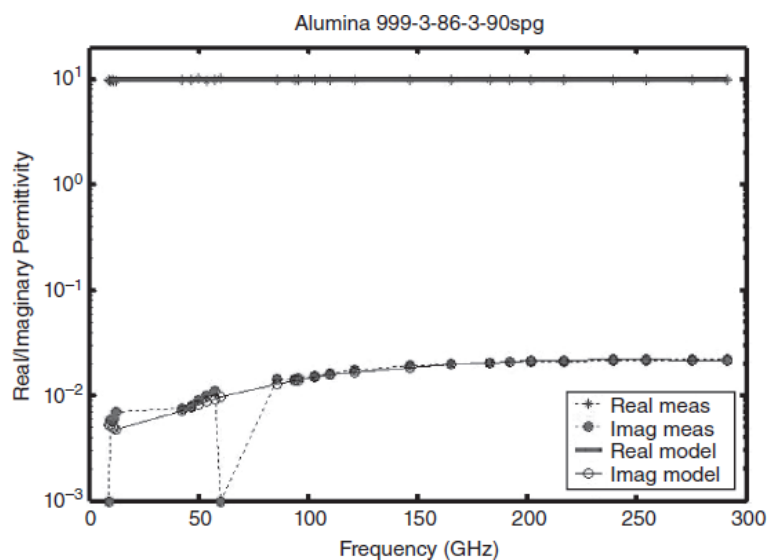


Figure 12.2 Measured and model fit of Alumina 999 permittivity.



Disclaimer: Data are results obtained by the author on samples prepared in the laboratory. Variations may be observed by other researchers when prepared in their laboratories. Data are not "guaranteed" by a manufacturer or the author. Variations in properties may be observed and users of the products should verify properties. All material properties are guides for engineering design. Materials prepared in the laboratory should be considered as experimental and also guides for engineering design.